

Introduction to General Relativity

Books Corral on GR

Great
reference
literacy

Schutz - 1st book on GR
"2nd edition"

Gravitation - Misner, Thorne, Wheeler
"Great reference"

GR by Wald "more math"

Definitions

SPACETIME: a manifold of
events endowed with a metric

Manifold: A set of points with
well understood connections
↓
read in
Corral 59-62
proportion connections

event: When and Where something
happens (coordinates is a label
for events)

But the event exists independently of labels

Metric: A notion of Distance
Between events in the
manifold $\rightarrow$ needed for the notion
of Distance in the manifold

Manifold

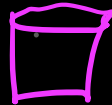
with
metric

without

$\rightarrow$ Metric

$\rightarrow$ Topologically

the same



$\rightarrow$ Geometry
is configured

SPECIAL Relativity

is the simplest Theory of spacetime

corresponds to General Relativity

with No Gravity

Inertial Reference frame

(following Bondi and H. Kip Thorne)

Lattice of clocks and measuring rods
That allows us to assign coordinates
to any event that happens in ST

Properties:

- Lattice moves freely through ST
(its INERTIAL)

↳ its at rest in respect to someone

- Measuring rods are orthogonal
to each other

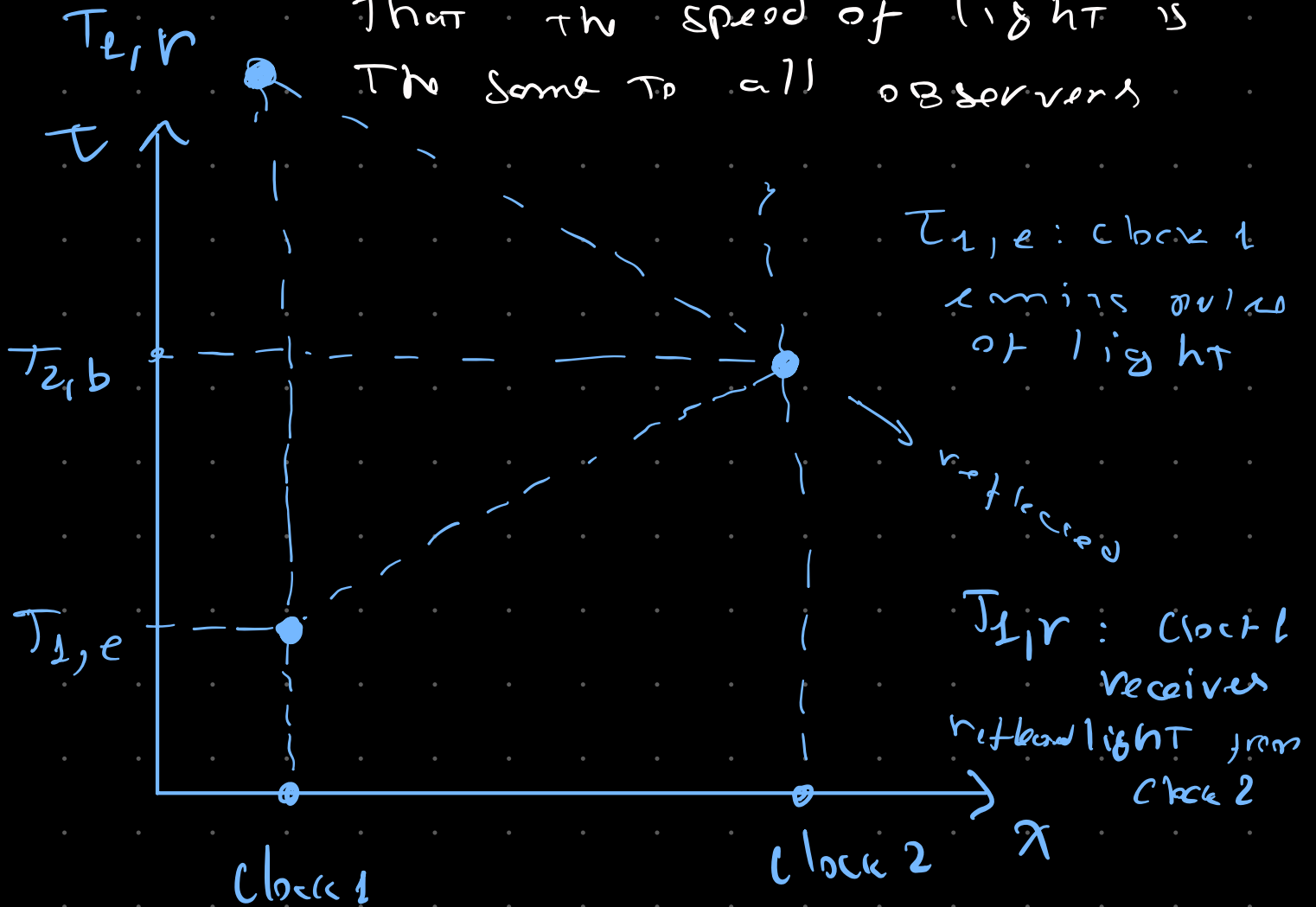
- rods
evenly marked 
in length

- Clocks tick uniformly

- all the clocks are in sync

With the "Einstein Sync Procedure"

↓
Takes advantage of the fact
that the speed of light is
the same to all observers



$$T_{2,b} = \frac{T_{1,e} + T_{1,r}}{2}$$

Units: Choose basic unit of length to
be distance light travels in your
basic unit of time

• If Basic unit is $s \rightarrow 1 \text{ light second}$

• If Basic unit is $1 \text{ ns} \rightarrow 1 \text{ light nanosecond}$
 $\swarrow$
 1 foot unit

• $c = \frac{1 \text{ light time unit}}{\text{Time unit}} = 1$

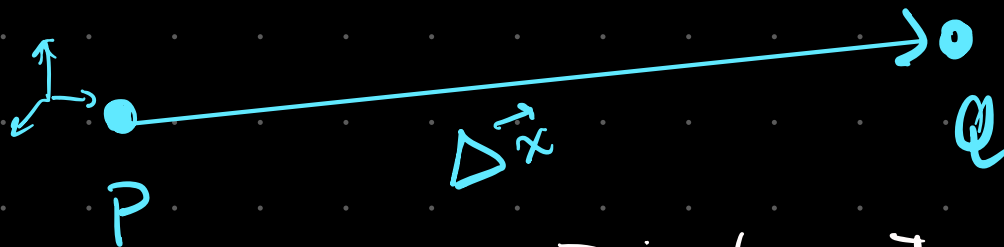
$\downarrow$
 So all velocities are dimensionless

Geometric OBJECT

$\bigcirc$ is an observer in the I.R.F. Special velocity for

Previously defined; $\bigcirc$ observe 2 events

P and Q



Displacement from P to Q

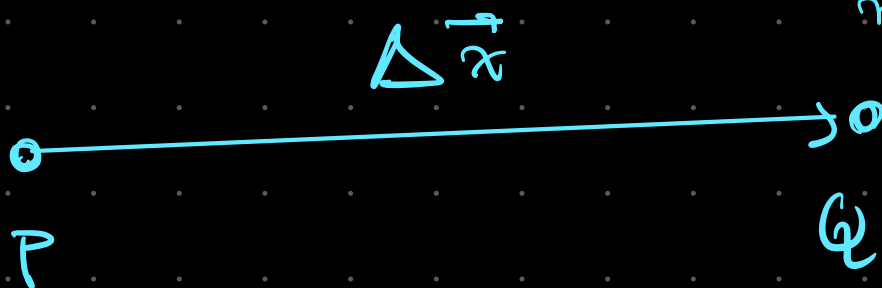
$$\Delta \vec{x} \stackrel{\circ}{=} (t_q - t_p, x_q - x_p, y_q - y_p, z_q - z_p)$$

$\swarrow$
 To observer $\bigcirc$

$$\Delta \vec{x} \xrightarrow{\mathcal{O}} \Delta x^\mu \quad \mu \in [t, x, y, z] \\ \in [0, 1, 2, 3]$$

Greek Indices tend to be used to label spacetime indices

In a particular Time use $\bar{\mathcal{O}}$ for a different Inertial Observer measured by $\bar{\mathcal{O}}$



$P, Q, \Delta \vec{x}$ are the same

Because they are Geometric Objects

that exists Independent of Representation

The Representation Does Change

$$\Delta \vec{x} \xrightarrow{\bar{\mathcal{O}}} \Delta x^{\bar{\mu}}$$

Lorentz Transformation relates $\mu \rightarrow \bar{\mu}$

$$\Delta x^{\bar{0}} = \gamma \Delta x^0 - \gamma v \Delta x^1$$

$$\Delta x^{\bar{1}} = -\gamma v \Delta x^0 + \gamma \Delta x^1$$

$$\Delta x^{\bar{2}} = \Delta x^2 \quad ; \quad \Delta x^{\bar{3}} = \Delta x^3$$

$\bar{G}$ moves with v along axis 1
with speed v as seen by G

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Better notation

$$\Delta \vec{x}^{\bar{\mu}} = \sum_{\nu=0}^3 \Lambda^{\bar{\mu}}_{\nu} \Delta x^{\nu}$$

$$\Delta x^{\bar{\mu}} = \Lambda^{\bar{\mu}}_{\nu} \Delta x^{\nu}$$

Einstein's Summation Convention:

Repeated indices in upstairs

and downstairs position are assumed

to be summed to their final index

In this case $\frac{3}{r}$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \Lambda^{\bar{\mu}}_0 \Delta x^0 + \Lambda^{\bar{\mu}}_1 \Delta x^1 \dots$$

$$\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \dots$$

$$\Lambda^{\bar{\mu}}_{\nu} \rightarrow \text{Lorentz transformation matrix} \quad \Lambda \rightarrow \text{Lorentz application}$$

$$\Lambda^{\bar{\mu}}_0 \rightarrow \text{for time}$$

Note:

$$\frac{\partial x^{\bar{\mu}}}{\partial x^{\nu}} = \Lambda^{\bar{\mu}}_{\nu} \quad \text{Trivial}$$

Notice: $\Delta x^{\bar{\mu}} = \Lambda^{\bar{\mu}}_{\nu} \Delta x^{\nu}$

$$\Delta x^{\bar{\mu}} = \Lambda^{\bar{\mu}}_{\mu} \Delta x^{\mu}$$

Calling this ν is not critical

$$\Lambda^{\bar{\mu}}_{\alpha} \Delta x^{\alpha}$$

So ν or α or μ is a
"Dummy Index"

$\bar{\mu}$ is NOT a Dummy Index
→ a "free Index"

SPACE TIME VECTOR

any quartet of numbers ("components")
which Transform Between $\pm n f's$
like Displacement vector

$$\vec{A} \stackrel{G}{=} (A^0, A^1, A^2, A^3) \xrightarrow{G} A^\alpha$$

$$\downarrow \quad \text{If } A^{\bar{\mu}} = \Lambda_{\alpha}^{\bar{\mu}} A^\alpha \quad \begin{array}{l} \text{If the } \uparrow \\ \text{Describes component} \\ \text{for } G \text{ from} \\ \text{ITS a vector} \end{array}$$

has to obey linearity laws

$$\vec{A}, \vec{B}$$

$$\leadsto \vec{C} = \vec{A} + \vec{B}$$

$$\leadsto n\vec{A} = \vec{D}$$

$n \rightarrow$ scalar invariant
for all observers
(the same)

Basis Vectors

In frame $\mathcal{O}$, write down 4 special vectors

$$\vec{e}_0 \doteq (1, 0, 0, 0) ; \vec{e}_1 \doteq (0, 1, 0, 0)$$

$$\vec{e}_2 \doteq (0, 0, 1, 0) ; \vec{e}_3 \doteq (0, 0, 0, 1)$$

Concise way of writing → Four components

$(\vec{e}_\alpha)^\beta$ reads
The β component of unit vector $\vec{e}_\alpha$

$$(\vec{e}_\alpha)^\beta \doteq \delta_\alpha^\beta \quad \text{Kronecker delta}$$

$$\begin{array}{l} \beta 0 \rightarrow \\ \beta 1 \rightarrow \\ \beta 2 \rightarrow \\ \beta 3 \rightarrow \end{array} \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right] \doteq (\vec{e}_\alpha)^\beta$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $\alpha 0 \quad \alpha 1 \quad \alpha 2 \quad \alpha 3$

Turning an $\vec{A}$ vector and the basis vector returns the true object,

$$\vec{A} = A^\alpha \vec{e}_\alpha$$

remember Einstein's summation

This is equivalent to

$$\vec{A} = \sum_{\beta=0}^3 A^{\alpha} \vec{e}_{\alpha}^{\beta}$$

$$\vec{A} = \begin{bmatrix} 1 \\ x \\ y \\ z \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ x \\ y \\ z \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} + \dots$$

↓

This is the geometric object (a vector)

How do basis vectors transform

$$\vec{A} = A^{\alpha} \vec{e}_{\alpha} = \bar{A}^{\bar{\mu}} \vec{e}_{\bar{\mu}}$$

$$= \left(\Lambda^{\bar{\mu}}_{\beta} A^{\beta} \right) \vec{e}_{\bar{\mu}}$$

II

o You can treat

This object like metric

BUT ITS gonna be impossible!

In the future

Exemplification for the Lorentz Transform Matrix

$$\Lambda^\beta_\alpha = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$A^\mu = \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix} \quad A^{\bar{\mu}} = \Lambda^\mu_\beta A^\beta$$

$$\begin{pmatrix} t' \\ x' \\ y' \\ z' \end{pmatrix} = \sum_{\mu=0}^3 \Lambda^\mu_\beta A^\beta$$

$$\Lambda^0_\beta A^\beta + \Lambda^1_\beta A^\beta + \Lambda^2_\beta A^\beta$$

$$\begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \end{pmatrix} \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix} + \begin{pmatrix} -\gamma\beta & \gamma & 0 & 0 \end{pmatrix} \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix} + \dots$$

fixing values

$$(\Lambda^0_0 A^0 + \Lambda^0_1 A^1 + \Lambda^0_2 A^2 + \Lambda^0_3 A^3) + \Lambda^1_0 A^0$$

$$(\gamma t - \gamma\beta x + 0 + 0)$$

Only (hit) $\rightarrow$ That's why
 we need the basis vector (vector)
 to make it a simple sum

$$(\Lambda_{\beta}^{\bar{\mu}} A^{\beta}) \vec{e}_{\bar{\mu}}$$

$$\begin{aligned}
 & (\Lambda_0^0 A^0 + \Lambda_1^0 A^1 + \Lambda_2^0 A^2 + \Lambda_3^0 A^3) \cdot \hat{t} \\
 & + (\Lambda_0^1 A^0 + \dots) \cdot \hat{x} + (\dots) \hat{y} + (\dots) \hat{z}
 \end{aligned}$$

its a vector

For this (can

$$(A^{\beta} \Lambda_{\beta}^{\bar{\mu}}) \vec{e}_{\bar{\mu}} = (\Lambda_{\beta}^{\bar{\mu}} A^{\beta}) \vec{e}_{\bar{\mu}}$$

Returning to class

$$(A^{\beta} \Lambda_{\beta}^{\bar{\mu}}) \vec{e}_{\bar{\mu}} \quad \text{as } \beta \text{ is a dummy index}$$

changing β has no consequences

$$A^{\alpha} \vec{e}_{\alpha} = A^{\beta} \Lambda_{\beta}^{\bar{\mu}} \vec{e}_{\bar{\mu}}$$

$$A^{\alpha} \vec{e}_{\alpha} = A^{\alpha} \Lambda_{\alpha}^{\bar{\mu}} \vec{e}_{\bar{\mu}}$$

$$\Lambda^\alpha (\vec{e}_\alpha - \Lambda^\mu_\alpha \vec{e}_\mu) = 0$$

This must be true for all vectors

means $\boxed{\vec{e}_\alpha = \Lambda^\mu_\alpha \vec{e}_\mu}$

Simple algorithm $\rightarrow$ "line up the indices"

Inverse Lorentz Transformation

(reverse velocity vector)

$$\vec{e}_\alpha = \Lambda^\mu_\alpha (v) \vec{e}_\mu$$

$$\vec{e}_\mu = \Lambda^\nu_\mu (-v) \vec{e}_\nu$$

$$\vec{e}_\alpha = \Lambda^\beta_\alpha (v) \vec{e}_\beta$$

$$\vec{e}_\alpha = \Lambda^\beta_\alpha (v) [\Lambda^\gamma_\beta (-v) \vec{e}_\gamma]$$

$$\vec{e}_\alpha = [\Lambda^\beta_\alpha (v) \Lambda^\gamma_\beta (-v)] \vec{e}_\gamma$$

requires Kronecker delta $\delta_\alpha^\gamma = \Lambda^\beta_\alpha (v) \Lambda^\gamma_\beta (-v)$

like with

$$\int \bar{\gamma} = \Lambda^{\beta}_{\alpha} \Lambda^{\bar{\gamma}}_{\beta}$$

Scalar Product Between 4-vectors

Recall that $\Delta s^2 = -\Delta t^2 + \Delta x^2 + \Delta y^2 + \Delta z^2$

Special relativity $\hookrightarrow$ its Invariant (Barnett)

$$\Delta s^2 \equiv \Delta \vec{x} \cdot \Delta \vec{x}$$

$$\stackrel{\cdot}{=} -(\Delta x^0)^2 + (\Delta x^1)^2 + (\Delta x^2)^2 + (\Delta x^3)^2$$

Since 4-vectors have the same Transformation Properties as $\Delta \vec{x}$ we similarly define

$$\vec{A} \cdot \vec{A} \stackrel{\cdot}{=} -(\bar{A}^0)^2 + (\bar{A}^1)^2 + (\bar{A}^2)^2 + (\bar{A}^3)^2$$

$\hookrightarrow$ must be Lorentz Invariant as well

Terminology

if $\vec{A} \cdot \vec{A} < 0 \rightarrow \vec{A}$ is a timelike vector

if $\vec{A} \cdot \vec{A} > 0 \rightarrow \vec{A}$ is a spacelike vector

$\vec{A} \cdot \vec{A} = 0 \rightarrow \vec{A}$ is light like or null

more general notation

$$\vec{A} \cdot \vec{B} = -A^0 B^0 + A^1 B^1 + A^2 B^2 + A^3 B^3$$

This is Invariant Define $\vec{C} = \vec{A} + \vec{B}$

$$\vec{C} \cdot \vec{C} = \vec{A} \cdot \vec{A} + \vec{B} \cdot \vec{B} + 2\vec{A} \cdot \vec{B}$$

Invariant

Invariant

Invariant



This must be Invariant

$$\vec{A} \cdot \vec{B} = (A^\alpha \vec{e}_\alpha) \cdot (B^\beta \vec{e}_\beta)$$

$$= A^\alpha B^\beta \vec{e}_\alpha \cdot \vec{e}_\beta$$

$$\vec{e}_\alpha \cdot \vec{e}_\beta = \eta_{\alpha\beta} \quad \text{Tensor}$$

Δs^2 this is like Pythagorean for ST

Flat ST

$$\vec{A} \cdot \vec{B} = A^\alpha B^\beta \cdot \eta_{\alpha\beta}$$

$$\eta_{\alpha\beta} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \rightarrow \text{The Metric Tensor}$$

$$\Delta s^2 = \vec{\Delta x} \cdot \vec{\Delta x} = \eta_{\alpha\beta} \Delta x^\alpha \Delta x^\beta$$

$$S, \quad \eta_{\alpha\beta} \Delta x^\alpha \Delta x^\beta \left\{ \begin{array}{l} > 0 \text{ more spatial } \Delta \\ < 0 \text{ more time } \Delta \\ = 0 \text{ a light beam connects them} \end{array} \right.$$

In Differential terms

$$ds^2 = d\vec{x} \cdot d\vec{x} = \eta_{\alpha\beta} dx^\alpha dx^\beta$$

$$d\vec{x} = dx^\alpha \vec{e}_\alpha$$

When this is true, $\vec{e}_\alpha$ is a coordinate basis vector

$\vec{e}_\alpha$ transforms a differential of your coordinate into a differential of a space time vector

Not interesting fact in Cartesian coordinates

In Curvilinear coordinates

$$dx = dx^{\hat{i}} \vec{e}_{\hat{i}}$$

$\hat{i} = 1$ radius
 $\hat{i} = 2$ theta
 $\hat{i} = 3$ phi

$$dx = dr \vec{e}_r + r d\theta \vec{e}_\theta + r \sin\theta d\phi \vec{e}_\phi$$

$$dr \vec{e}_r + r d\theta \vec{e}_\theta + r \sin\theta d\phi \vec{e}_\phi$$

$\vec{e}_{\hat{i}}$ is an orthonormal basis

$$\vec{e}_i \cdot \vec{e}_j = \delta_{ij}$$

not necessarily
in case

$$\vec{e}_r \cdot \vec{e}_r = 1 ; \vec{e}_\theta \cdot \vec{e}_\theta = r^2 ; \vec{e}_\phi \cdot \vec{e}_\phi = r^2 \sin^2 \theta$$

if 2th norm is

$$\vec{e}_i \vec{e}_j = \eta_{ij} = \delta_{ij}$$

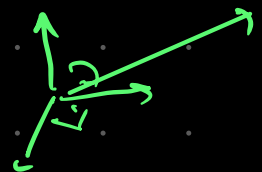
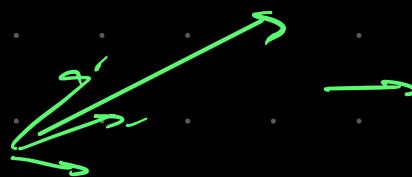
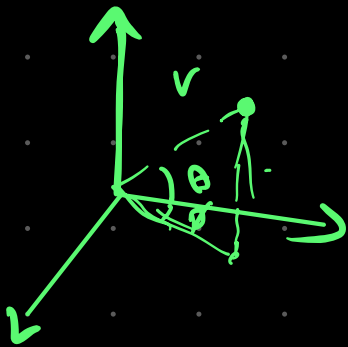
if not

$$\vec{e}_i \vec{e}_j = \eta_{ij}$$

Understanding

$$\rightarrow dr \vec{e}_r + r d\theta \vec{e}_\theta + r \sin \theta d\phi \vec{e}_\phi$$

$$dr \vec{e}_r + d\theta \vec{e}_\theta + d\phi \vec{e}_\phi$$



Important 4 vectors

$$\vec{u} = \frac{d\vec{x}}{d\tau} \rightarrow \text{4 velocity}$$

$d\tau$ = Time Interval
as measured
along the trajectory
of the observer
with velocity $\vec{u}$
(proper time)

$$\vec{u} = \begin{pmatrix} \gamma \\ \gamma v \end{pmatrix}$$

In the rest frame of
the observer with velocity $\vec{u}$

$$\vec{u} = (1, 0)$$

4-momentum

$$\vec{p} = m \vec{u}$$

$\hookrightarrow$ rest mass of the object

$$\vec{p} = (\gamma m, \gamma m v) \quad \text{for} \quad \hookrightarrow \text{relativistic}$$

$$\vec{p} = (E, \vec{p})$$

Couple them to scalar

$$\left\{ \vec{u} \cdot \vec{u} = -\gamma^4 + \gamma^2 v^2 = -1 \right\}$$

$$\vec{p} \cdot \vec{p} = m^2 \vec{u} \cdot \vec{u} = -m^2 = -E^2 + |\vec{p}|^2$$

$$-m^2 = -E^2 + |\vec{p}|^2 \quad \underline{c=1}$$

$$E^2 = m^2 + |\vec{p}|^2$$

$$E^2 = m^2 c^4 + |\vec{p}|^2 c^2$$

$$E = mc^2$$